

Multiple Choice (10 marks)

1. The theory of evolution is most accurately described as
 - (a) a universally accepted fact about the origin and development of all species.
 - (b) a hypothesis in the process of being tested and verified.
 - (c) a disproven theory about the development of species over time.
 - (d) a speculation about possible changes in populations over time.
 - (e) an opinion that some scientists hold about how living things change over time.
2. Why is genetic drift less important in human evolution at present than it was twenty thousand or more years ago?
 - (a) The human genome has fully adapted to current environmental pressures
 - (b) Modern medicine has eliminated the effects of genetic drift
 - (c) The rate of genetic drift is directly proportional to technological advancements
 - (d) Humans have reached a genetic equilibrium where drift is no longer possible
 - (e) Humans have developed immunity to genetic drift
3. A male bacterium conjugates with a female bacterium. After conjugation, the female becomes a male. Account for this "sexchange".
 - (a) The female bacterium becomes a male as a result of nutrient deficiency
 - (b) The female becomes a male by receiving one copy of the F factor
 - (c) The female bacterium becomes a male through the process of osmosis
 - (d) The female bacterium becomes a male through gene mutation
 - (e) The female bacterium becomes a male after receiving multiple copies of non-F factor genes
4. To prevent desiccation and injury, the embryos of terrestrial vertebrates are encased within a fluid secreted by the
 - (a) blastoderm
 - (b) mesoderm
 - (c) amnion
 - (d) zona pellucida
 - (e) ectoderm
5. Why are the stem reptiles especially important in evolution?

- (a) They exclusively evolved into modern reptiles without any other descendants
- (b) They were the first reptiles to return to an aquatic lifestyle
- (c) They gave rise to only the dinosaurs
- (d) They were the only reptiles to develop flight
- (e) They existed only in the Mesozoic Era

True / False (10 marks)

Answer True or False

- 6. True or False: Sister chromatids separate during meiosis II, not meiosis I, which involves the separation of homologous chromosomes.
- 7. True or False: The acrosome and cortical reactions allow multiple sperm to fertilize the egg, leading to the formation of twins.
- 8. True or False: The drop in blood pressure can lead to blood clotting due to reduced shear stress on endothelial cells.
- 9. True or False: An extra Y chromosome (XXY) is linked to an increase in cognitive abilities and intelligence.
- 10. True or False: Xylem vessel members undergo programmed cell death to facilitate water transport in plants.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms by which the pharynx directs food to the esophagus and air to the trachea during swallowing, emphasizing the roles of the epiglottis and neural reflexes.
- 12. Discuss the changes that occur in the A band of a sarcomere during muscle contraction, and analyze how these changes contribute to the overall process of muscle shortening.

End of Paper